

1. Select Requirement Gathering Techniques

Technique 1: Interviews

Interviewing a dorm proctor and security guard to confirm real workflows; how rooms are assigned, how penalties are recorded, and how laundry exits are managed.

Interviews provide deep, qualitative insights.

Technique 2: Survey

Surveying students to priorities features. For example, how often do they need maintenance, whether they want a QR-based laundry system.

Surveys reach many stakeholders quickly.

Technique 3: Document Analysis

Analyzing existing documents: complaint registers, penalty logbooks, room allocation sheets, and duty rosters.

This

helps in understanding the current process and identify gaps.

Technique 4: Brainstorming

In a team workshop, we will brainstorm “blue-sky” ideas such as automatic room assignments, push notifications to staff, or a penalty appeal system.

Brainstorming encourages creative requirements.

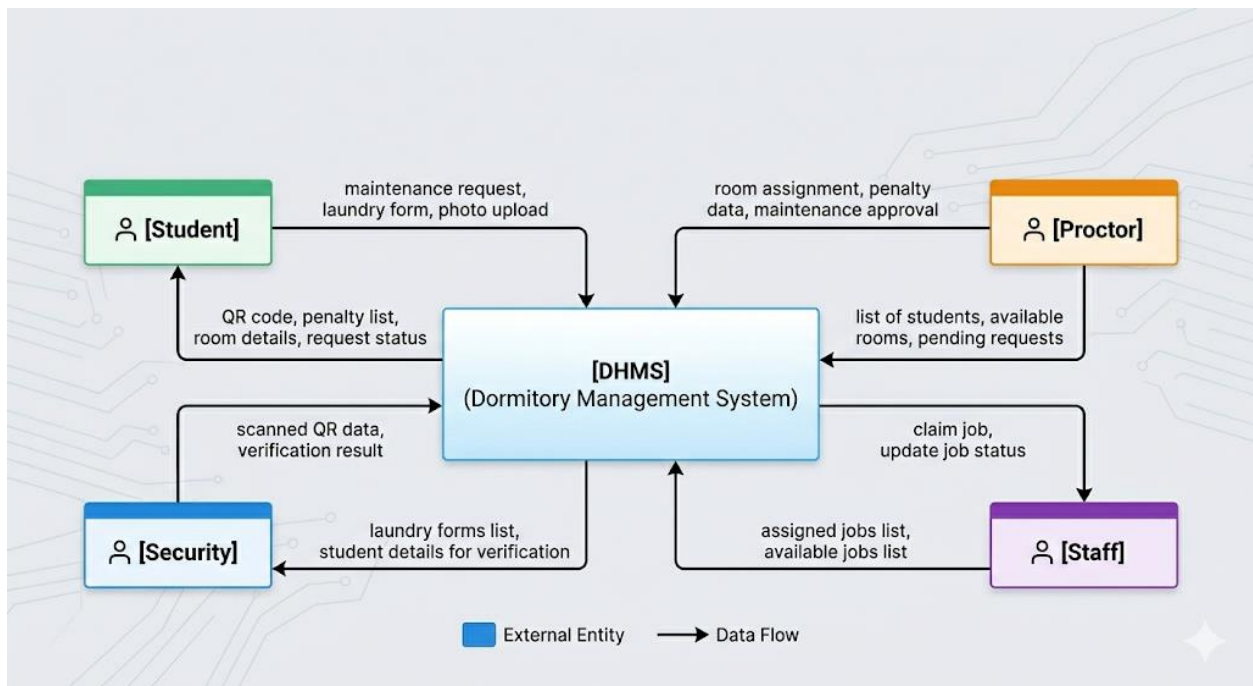
2. Essential Use Case Modeling

1. Use Case Summary

Use Case Name	Primary Actor	Description (user goal)
Assign Room to Student	Proctor	The proctor selects a student, a dormitory, a room number, an assignment date, and an expected check-out date.
View Available Rooms	Proctor	The proctor views a list of all rooms with details (dorm, floor, type, capacity, occupancy, status).
Assign Penalty	Proctor	The proctor selects a student, a violation type, description, duration, and start date; the system calculates the end date.
Approve Maintenance Request	Proctor	The proctor reviews pending requests and assigns them to appropriate staff.
Submit Maintenance Request	Student	The student describes an issue (type, title, description) and sets an urgency level.
Submit Laundry Form	Student	The student enters number of items, item list, and special instructions; the system generates a QR code for exit.
Verify Laundry Forms	Security	The security guard views a list of submitted laundry forms and marks them as verified.

Scan Laundry QR Code	Security	The security guard scans the student's QR code using a camera to verify laundry exit.
View My Assigned Jobs	Staff	The staff member sees all jobs assigned to them (ID, room, issue, urgency, status).
Claim Available Job	Staff	The staff member browses unassigned jobs and claims one matching their skill.

2. Context Diagram



3. Primary and Secondary Actors

Actor	Type	Description
Student	Primary	A resident who uses the system to request maintenance, submit laundry forms, and view penalties.
Proctor	Primary	A dormitory supervisor who assigns rooms, approves maintenance requests, assigns penalties, and views student lists.
Staff	Primary	Technical staff who receive assigned jobs, update job status, and claim available tasks.
Security Guard	Primary	A guard who verifies laundry exit QR codes and manages student check-in/out.
Notification Service	Secondary	An external system (email/SMS) that sends alerts to students and staff.

4. Requirements Anticipation, Elicitation, Assurance, Specification

A. Requirements Anticipation: Predicting what users will need without being explicitly asked, based on domain knowledge and experience.

We anticipate that students will want to track the status of their maintenance request (e.g., “pending → assigned → completed”) even though no student has mentioned it yet. We will include a status field in the requirements.

B. Requirements Elicitation: Actively gathering requirements from stakeholders through interviews, surveys, workshops, observation, etc.

We will elicit requirements by showing a mock-up of the penalty form to a proctor and asking: “What violation types should be included? Do you need a ‘consequences’ field?”.

C. Requirements Assurance: Ensuring that requirements are correct, consistent, feasible, and complete by using validation techniques such as inspections, walkthroughs, and prototyping.

We will perform assurance by reviewing all requirements with our team and a proctor. We will check that “the system shall generate a QR code for laundry” is consistent with “security shall scan QR code” and is feasible using open-source libraries.

D. Requirements Specification: Writing requirements clearly and unambiguously in a document (e.g., Software Requirements Specification).

Example specification: “**The DHMS shall allow a student to submit a maintenance request by providing issue type (plumbing, electrical, furniture, other), a brief title, a detailed description, and an urgency level (low, medium, high).**” This will be documented in our SRS.